A++ / HIGH WEIGHTAGE (Must Do First)

1. Molecular Basis of Inheritance

[2024, 2023, 2019, 2018, 2016] 5 Markers: Describe the Lac Operon in E. coli with the help of a schematic diagram. How does it switch on and off?
[2023, 2020, 2017, 2014] 5 Markers: Describe the Hershey-Chase experiment. Write the conclusion drawn by the scientists.
[2024, 2019, 2016, 2015] 3 Markers: Explain the process of transcription in prokaryotes. How is it different from eukaryotes?
[2022, 2020, 2018] 3 Markers: State the salient features of the genetic code.
[2023, 2018, 2016, 2012] 2/3 Markers: Explain Meselson and Stahl's experiment to prove that DNA replication is semi-conservative.

2. Principles of Inheritance & Variation

[2024, 2022, 2020, 2017] Case-Based/5 Markers: A color-blind man marries a woman with normal sight who has no history of color blindness in her family. What is the probability of their grandson being color-blind? (Show the cross).
[2023, 2019, 2014, 2013] 3 Markers: Differentiate between incomplete dominance and codominance with a suitable example for each.
[2024, 2020, 2018, 2017] 3 Markers: What is Down's syndrome? State its chromosomal basis, cause, and any three symptoms.
[2023, 2019, 2015] 2 Markers: Explain the mechanism of sex determination in birds.

3. Human Reproduction

[2024, 2022, 2019, 2016] 5 Markers: Explain the menstrual cycle in human females. Describe the hormonal regulation of this cycle.
[2023, 2020, 2018, 2015] 5 Markers: Draw a neat, labeled diagram of a sectional view of the human male reproductive system OR a sectional view of the seminiferous tubule.
[2024, 2019, 2017, 2014] 3 Markers: Describe the process of spermatogenesis with the help of a schematic flow chart.
[2022, 2018, 2016] 3 Markers: Where does fertilization occur in humans? Explain the events that occur during this process.

4. Sexual Reproduction in Flowering Plants

[2024, 2023, 2020, 2019, 2017] 5 Markers: Explain the process of double fertilization in angiosperms. Why is it called "double" fertilization?
[2022, 2018, 2016, 2015] 3 Markers: Draw a labeled diagram of a typical anatropous ovule OR a mature embryo sac.
[2023, 2019, 2015] 3 Markers: Differentiate between microsporogenesis and megasporogenesis. Which type of cell division occurs during these events?
[2024, 2020, 2017] 2 Markers: What is outbreeding, and what are the devices plants use to promote it?

A / HIGH-MODERATE WEIGHTAGE

5. Biotechnology: Principles & Processes

[2024, 2022, 2020, 2017] 5 Markers: Describe the steps involved in Recombinant DNA technology.
[2023, 2019, 2016, 2014] 3 Markers: What are restriction endonucleases? Explain their functioning with the help of an example (like EcoRI).
[2024, 2018, 2016, 2013] 3 Markers: Explain the process of Polymerase Chain Reaction (PCR) with a diagram.
[2023, 2020, 2019, 2017] 2 Markers: What is the principle behind Gel Electrophoresis?

6. Biotechnology & Applications

[2024, 2022, 2020, 2019, 2016] 3/5 Markers: How has biotechnology been used to produce nematode-resistant tobacco plants? Explain the process of RNA interference (RNAi).
[2023, 2020, 2018, 2016] 3 Markers: How is Genetically Engineered insulin (Humulin) produced? What was the main challenge in producing it, and how did Eli Lilly overcome it?
[2024, 2019, 2017, 2015] 3 Markers: What are Cry proteins? Name the organism that produces it. How has man exploited this protein to his benefit (Bt Cotton)?

7. Biology in Human Welfare (Human Health & Disease)

[2024, 2020, 2019, 2017, 2014] 5 Markers: Trace the life cycle of Plasmodium in the human body and the mosquito with the help of a flow chart.
[2023, 2019, 2018, 2015] 3 Markers: Draw a well-labeled diagram of an antibody molecule.
[2024, 2022, 2017] 3 Markers: Differentiate between active and passive immunity with examples.
[2023, 2020, 2016] 3 Markers: Explain the mechanism by which HIV replicates inside the host cell.

8. Evolution

[2024, 2023, 2019, 2016] 3/5 Markers: State the Hardy-Weinberg principle. Write the algebraic equation and explain any three factors that affect the Hardy-Weinberg equilibrium.
[2022, 2020, 2018, 2015] 3 Markers: Differentiate between homologous and analogous organs with examples. How do they support the theory of evolution (divergent vs. convergent)?
[2024, 2019, 2017, 2015] 2 Markers: Describe Miller's experiment with a labeled diagram.

B / MODERATE WEIGHTAGE

9. Reproductive Health

[2023, 2019, 2016] 2/3 Markers: What is amniocentesis? Why is there a statutory ban on it?
[2024, 2020, 2017, 2014] 3 Markers: Suggest and explain any three Assisted Reproductive Technologies (ART) to help infertile couples (e.g., IVF, ZIFT, GIFT).

10. Microbes in Human Welfare

[2024, 2022, 2019, 2018, 2016] 3 Markers: Explain the role of microbes in the secondary treatment of sewage. How does BOD relate to it?
[2023, 2020, 2017, 2015] 2 Markers: Match the following microbes to their bioactive molecules.

11. Organisms & Populations

[2024, 2023, 2020, 2019] 3 Markers: Explain mutualism, parasitism, and commensalism with one example each.
[2022, 2018, 2016, 2014] 3 Markers: Differentiate between exponential and logistic growth models. Which one is more realistic and why?

C / LOW WEIGHTAGE (Do Last)

12 & 13. Ecosystem & Biodiversity

[2024, 2020, 2018, 2016] 3 Markers: Explain the "Evil Quartet" (the four major causes of biodiversity loss).
[2023, 2019, 2017, 2015] 2 Markers: Differentiate between in-situ and ex-situ conservation with examples.
[2022, 2019, 2016] 2 Markers: Explain the 10% law of energy transfer in a food chain.